

Gustavo Rodriguez

Mechanical Engineering Student | Expected Graduation: December 2028
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Research & Engineering Leadership

IEEE EXO, University of South Florida

Project Lead / EXO Chair and President

Tampa, FL

August 2025 – Present

- Lead a multidisciplinary lower-limb exoskeleton program spanning mechanical design, electrical, controls, research, fabrication, testing, safety, procurement, and technical documentation.
- Define requirements for a unilateral hip/knee assistive prototype using AK80-9 actuators, IMUs, encoders, Jetson Orin Nano, and Teensy 4.1; perform OpenSim gait comparisons across healthy, hemiparetic, and SHIA datasets to support assistance timing, torque targets, and validation planning.

CARRT Lab, University of South Florida

REU Research Assistant

Tampa, FL

August 2026 – Present

- Analyze IMU and Vicon motion-capture data and support experimental setup, participant data collection, sensor validation, post-processing, data-quality review, and technical documentation for human-motion research.
- Develop a modular ESP32/IMU sensor for rugby lineout motion capture; designed a SolidWorks ankle/thigh enclosure reduced from 80 x 60 mm to 43 x 32 mm (71.3% smaller footprint), with ventilation and integrated adjustable strap slots to avoid reliance on athletic tape or sleeves. Helped validate the PCB and supported compact PCB/layout integration.

Engineering Experience

Lonza

Reliability Engineering Intern

Tampa, FL

August 2026 – December 2026

- Investigate recurring Xcelodose 600S and microbalance failures through physical inspection, mechanical alignment, spring/load-cell geometry, corner-load behavior, SOP-to-practice comparison, 6M, Gemba, and failure-mode analysis.

Lonza

Facilities, Engineering & Maintenance Intern

Tampa, FL

May 2026 – August 2026

- Led a Lean/5S asset-identification improvement that reduced representative equipment search time from about 9 minutes to 1.5 minutes (83%).
- Built Power Apps/Power Automate/SharePoint workflows for maintenance and manufacturing traceability, including guided request intake and structured run/status tracking.

Technical Skills & Credentials

Engineering: SolidWorks 2025, CAD/assembly modeling, drawing/specification review, FDM 3D printing, RCA/FMEA, equipment validation

Software / Hardware: MATLAB, OpenSim, Python, Excel, Power Apps, Power Automate, SharePoint, Arduino/ESP32, Jetson/Teensy, IMUs, sensors and actuators

PCB / Electronics: Altium PCB Design coursework and Altium Education certificate in progress; IEEE Altium PCB design workshops; PCB validation/integration support

Credentials: Lean Six Sigma Yellow Belt; CSWA CAD Design; CSWA Sustainability; ReCF Pre-Engineering & Robotics; CITI Biomedical / Research Integrity

Selected Technical Projects

SMURF – Maintenance Request System | Power Apps, Power Automate, SharePoint

- Built a two-app maintenance workflow with structured condition questions, automated urgency determination, HTML notifications/reminders, assignment/status tracking, and searchable records.

Mime Robotic Hand | Python, MediaPipe, ESP32, PCA9685, Servo Control

- Integrated computer vision with an open-source 5-DOF tendon-driven hand; developed calibration, mapping, smoothing/hysteresis, reverse-servo logic, and interactive control modes.

Education

University of South Florida

B.S. Mechanical Engineering, GPA: 3.33

Expected December 2028

Coursework: Dynamics, Mechanics of Solids, Thermodynamics, Materials Engineering, Electrical Systems, CAD & Engineering, PCB Design / Altium.